

HDOC DAY-24 ASSIGNMENT

Increasing Patterns Using Switch-Case and Nested Loops

1. Problem Understanding

The task is to write a menu-driven C program using a switch statement to print the “Increasing patterns” selected by the user. Nested loops are used without arrays. Each pattern is implemented in a separate parameterless function with no return value.

For each activity, prepare the algorithm and test-case table. Then write, compile and execute the C program, and test it against the prepared test cases.

- Increasing right-angled triangle
- Increasing row-number triangle
- Increasing natural-number pattern
- Consecutive-number triangle
- Increasing alphabet pattern

2. Standard Test Cases

Test Case	Choice	Input (n)	Expected Output	Status
TC1	1	4	Increasing right-angled triangle with 4 rows	Pass
TC2	1	3	Increasing right-angled triangle with 3 rows	Pass
TC3	2	4	Increasing row-number triangle with 4 rows	Pass
TC4	2	3	Increasing row-number triangle with 3 rows	Pass
TC5	3	4	Increasing natural-number pattern with 4 rows	Pass
TC6	3	3	Increasing natural-number pattern with 3 rows	Pass
TC7	4	4	Consecutive-number triangle with 4 rows	Pass
TC8	4	3	Consecutive-number triangle with 3 rows	Pass
TC9	5	4	Increasing alphabet	Pass

			pattern with 4 rows	
TC10	5	3	Increasing alphabet pattern with 3 rows	Pass

3. Algorithm

A. Common Menu Logic

1. Start.
2. Display the menu options.
3. Read the user's choice.
4. Use switch-case to select the required pattern.
5. Call the corresponding void function.
6. The selected function reads the number of rows n and prints the required pattern using nested loops.
7. If the choice is invalid, display "Invalid choice".
8. Stop.

B. Increasing Right-Angled Triangle

9. Start.
10. Read the number of rows n.
11. Use an outer loop from 1 to n for the rows.
12. Use an inner loop from 1 to the current row number.
13. Print an asterisk followed by a space.
14. After each row, move to the next line.
15. Repeat until all n rows are printed.
16. Stop.

C. Increasing Row-Number Triangle

17. Start.
18. Read the number of rows n.
19. Use an outer loop from 1 to n.
20. Use an inner loop from 1 to the current row number.
21. Print the current row number followed by a space.
22. After each row, move to the next line.
23. Repeat until all n rows are printed.
24. Stop.

D. Increasing Natural-Number Pattern

25. Start.
26. Read the number of rows n.
27. Use an outer loop from 1 to n.
28. Use an inner loop from 1 to the current row number.
29. Print the inner loop number followed by a space.

30. After each row, move to the next line.
31. Repeat until all n rows are printed.
32. Stop.

E. Consecutive-Number Triangle

33. Start.
34. Read the number of rows n.
35. Set num = 1.
36. Use an outer loop from 1 to n.
37. Use an inner loop from 1 to the current row number.
38. Print num followed by a space and increase num by 1.
39. After each row, move to the next line.
40. Repeat until all n rows are printed.
41. Stop.

F. Increasing Alphabet Pattern

42. Start.
43. Read the number of rows n.
44. Use an outer loop from 1 to n.
45. Use an inner loop from 1 to the current row number.
46. Print the character 'A' + j - 1 followed by a space.
47. After each row, move to the next line.
48. Repeat until all n rows are printed.
49. Stop.

4. C Program

```
#include <stdio.h>
```

```
void rightTriangle()
```

```
{
```

```
    int n, i, j;
```

```
    printf("Enter number of rows: ");
```

```
    scanf("%d", &n);
```

```
    for (i = 1; i <= n; i++)
```

```
    {
```

```
        for (j = 1; j <= i; j++)
```

```
        {
```

```
            printf("* ");
```

```

    }

    printf("\n");
}

}

void rowNumberTriangle()
{
    int n, i, j;

    printf("Enter number of rows: ");

    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= i; j++)

        {
            printf("%d ", i);

        }

        printf("\n");
    }
}

void naturalNumberPattern()
{
    int n, i, j;

    printf("Enter number of rows: ");

    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= i; j++)

        {
            printf("%d ", j);

        }

        printf("\n");
    }
}

```

```

    }
}

void consecutiveNumberTriangle()
{
    int n, i, j, num = 1;

    printf("Enter number of rows: ");

    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= i; j++)
        {
            printf("%d ", num);

            num++;
        }

        printf("\n");
    }
}

void alphabetPattern()
{
    int n, i, j;

    printf("Enter number of rows (1 to 26): ");

    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= i; j++)
        {
            printf("%c ", 'A' + j - 1);
        }

        printf("\n");
    }
}

```

```
}  
  
int main()  
{  
    int choice;  
  
    printf("MENU\n");  
  
    printf("1. Increasing right-angled triangle\n");  
    printf("2. Increasing row-number triangle\n");  
    printf("3. Increasing natural-number pattern\n");  
    printf("4. Consecutive-number triangle\n");  
    printf("5. Increasing alphabet pattern\n");  
  
    printf("Enter your choice: ");  
  
    scanf("%d", &choice);  
  
    switch (choice)  
    {  
        case 1:  
            rightTriangle();  
            break;  
        case 2:  
            rowNumberTriangle();  
            break;  
        case 3:  
            naturalNumberPattern();  
            break;  
        case 4:  
            consecutiveNumberTriangle();  
            break;  
        case 5:  
            alphabetPattern();  
            break;  
        default:
```

```

printf("Invalid choice\n");

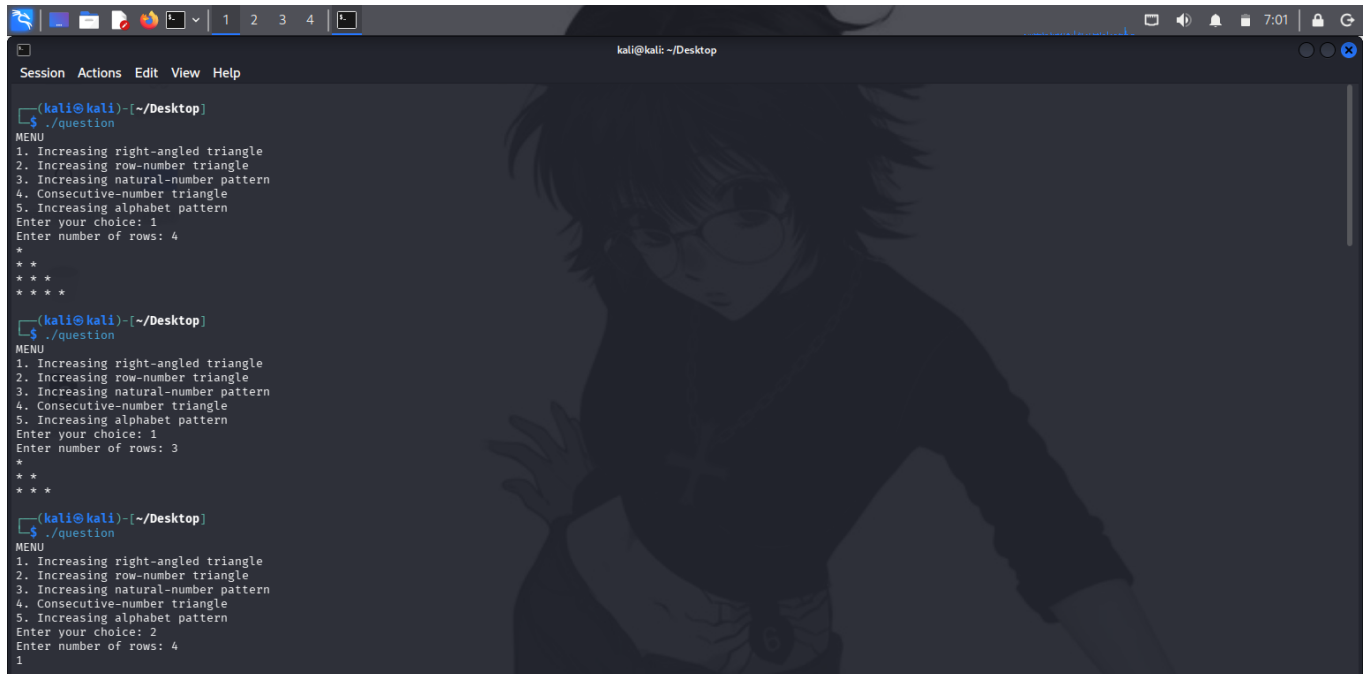
}

return 0;

}

```

5. Compilation and Execution Evidence



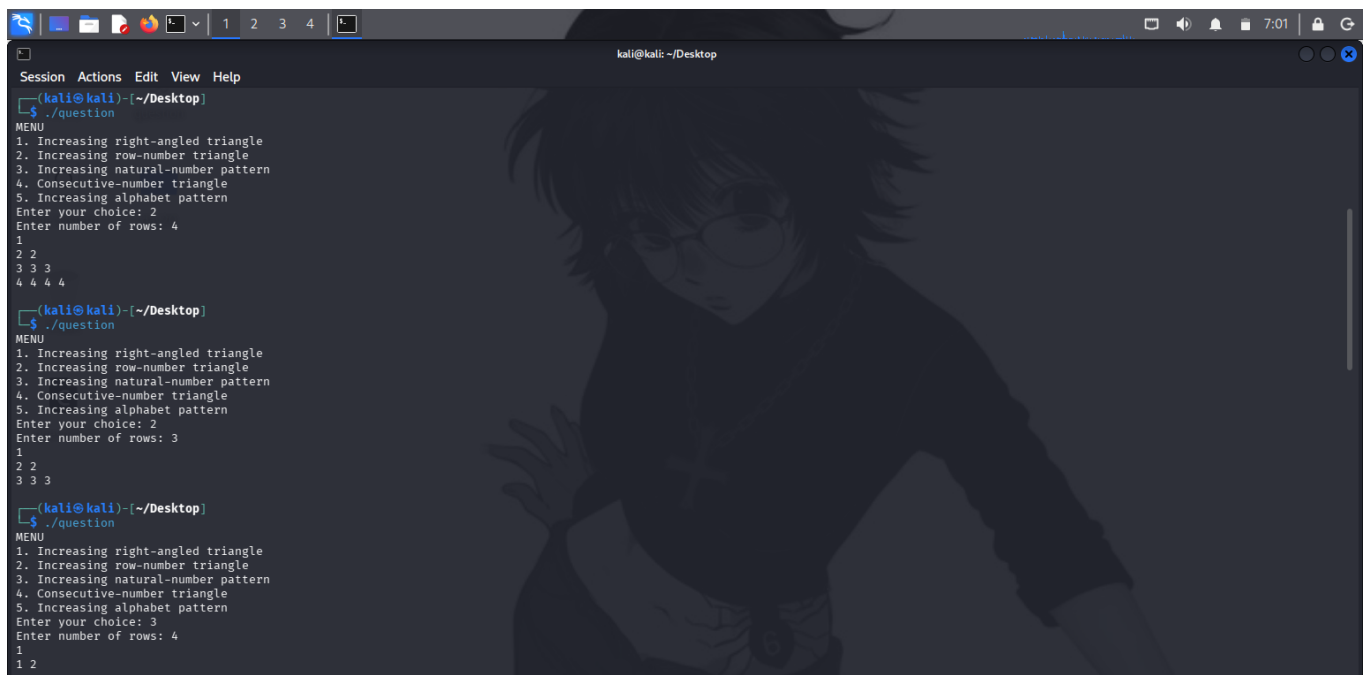
```

kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)~/Desktop
$ ./question
MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 1
Enter number of rows: 4
*
* *
* * *
* * * *

(kali@kali)~/Desktop
$ ./question
MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 1
Enter number of rows: 3
*
* *
* * *

(kali@kali)~/Desktop
$ ./question
MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 2
Enter number of rows: 4
1
2 2
3 3 3
4 4 4 4

```



```

kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)~/Desktop
$ ./question
MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 2
Enter number of rows: 4
1
2 2
3 3 3
4 4 4 4

(kali@kali)~/Desktop
$ ./question
MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 2
Enter number of rows: 3
1
2 2
3 3 3

(kali@kali)~/Desktop
$ ./question
MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 3
Enter number of rows: 4
1
1 2

```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
3 3 3

(kali@kali)~/Desktop
$ ./question
MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 3
Enter number of rows: 4
1
1 2
1 2 3
1 2 3 4

(kali@kali)~/Desktop
$ ./question
MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 3
Enter number of rows: 3
1
1 2
1 2 3

(kali@kali)~/Desktop
$ ./question
MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 4
Enter number of rows: 4
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 4
Enter number of rows: 3
1
2 3
4 5 6

(kali@kali)~/Desktop
$ ./question
MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 5
Enter number of rows (1 to 26): 4
A
A B
A B C
A B C D

(kali@kali)~/Desktop
$ ./question
MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 5
Enter number of rows (1 to 26): 3
A
A B
A B C

(kali@kali)~/Desktop
$
```

6. Verification of All Test Cases

Test Case	Choice	Input (n)	Expected Result	Actual Result	Status
TC1	1	4	4-row right-angled triangle	4-row right-angled triangle	Pass
TC2	1	3	3-row right-angled triangle	3-row right-angled triangle	Pass
TC3	2	4	4-row row-number	4-row row-number	Pass

			triangle	triangle	
TC4	2	3	3-row row-number triangle	3-row row-number triangle	Pass
TC5	3	4	4-row natural-number pattern	4-row natural-number pattern	Pass
TC6	3	3	3-row natural-number pattern	3-row natural-number pattern	Pass
TC7	4	4	1 to 10 in triangular pattern	1 to 10 in triangular pattern	Pass
TC8	4	3	1 to 6 in triangular pattern	1 to 6 in triangular pattern	Pass
TC9	5	4	A, A B, A B C, A B C D	A, A B, A B C, A B C D	Pass
TC10	5	3	A, A B, A B C	A, A B, A B C	Pass